

JONATHAN NADEAU

DATA ANALYST

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Portfolio Website: nadeaujonny.github.io

SUMMARY

Mathematics graduate (UCSB) and Google Data Analytics certified analyst, sole developer of four internal data tools now in use at a defense and telecom electronics manufacturer. Work spans legacy-system extraction, Python ETL and application development, live vendor API integration, and cost and availability analysis, with results including 4,038 hardware failure records made searchable, 21 datasheet-confirmed BOM spec errors surfaced, and roughly \$26K cut from a 100-board parts plan. Backed by 1,500+ hours communicating quantitative findings to non-technical audiences.

TECHNICAL SKILLS

Python: pandas, NumPy, scikit-learn, openpyxl, requests, pytest, Streamlit, Tkinter, PyInstaller, regex

Data Engineering & APIs: ETL pipeline design, REST and OAuth2 integration, TTL caching, rate limiting, incremental refresh, data-quality auditing and validation

SQL & Databases: Oracle SQL, BigQuery, MySQL, SQLite; CTEs, window functions, stored procedures, DDL/DML

Spreadsheets & BI: Excel (Power Query, Power Pivot, DAX, dynamic arrays, Solver), Power BI, Tableau, Looker Studio; R (tidyverse)

Engineering Practice: pytest suites, defect-injection testing, Git/GitHub, GitHub Actions, Windows packaging and deployment, runbooks and user guides

PROFESSIONAL EXPERIENCE

Data Analyst & Application Developer Intern | Escape Communications

July 2026 – Present | Torrance, CA

Sole developer of four internal tools for the engineering and procurement functions of a defense and telecom hardware manufacturer; owned requirements, architecture, implementation, validation, documentation, and deployment on each.

- **Failure Analysis Search Tool:** Reverse-engineered an undocumented legacy Bugzilla instance and extracted the full 4,038-record hardware failure corpus (2007–2026, 13,252 comments) via REST API; a full-corpus quality audit proved the structured failure fields unusable as signal (80%+ blank or defaulted) and redirected the design toward the comment threads where the engineering knowledge lived.
- Built a deterministic Python pipeline indexing 2,245 PCB reference designators out of free text, hand-validated per prefix by stratified sampling, eliminating a 68–71% false-positive rate on the worst values; shipped a Streamlit app with component-level lookup, TF-IDF similarity search, and watermark-based incremental refresh, with runbook, user guide, and build log enabling zero-developer operation.
- **BOM Checker (shipped company-wide):** Designed and shipped a standalone Windows tool (Python, pandas, openpyxl, PyInstaller) flagging 16 classes of structural and parametric defect in raw PADS Logic BOM exports without modifying a single cell, a flag-don't-fix design set in working sessions with the VP of Engineering; 427 tests including 26 defect-injection cases, output fidelity verified cell by cell at zero value and number-format differences.
- Built an offline decoder for 10+ manufacturers' part-numbering schemes, cross-checking BOM-stated specs against what each part number encodes; across 8 production BOMs spanning 5 product lines (2,332 rows) it surfaced 21 datasheet-confirmed spec errors, including a 100V requirement on a 16V-rated part, and 13 cross-board spec contradictions.
- **BOM Sourcing & Purchase Planning Tool:** Built an end-to-end sourcing pipeline (703 passing tests) that validates 300+ line BOM exports, prices and risk-assesses every part against the live Mouser and DigiKey (OAuth2) APIs, and outputs an auditable multi-vendor buy plan covering obsolescence risk, cost, and restock timing. Volume price breaks invisible to the prior manual process cut a real 100-board parts plan from \$129.8K to \$103.8K, roughly 20%, and availability checks flagged 55 at-risk parts ahead of the build.
- Diagnosed a suspected vendor-API reliability problem as a packaging-variant selection defect by measuring 266 rows against both vendors' own tools, raising priced coverage from 65 to 195 rows; built the API layer with TTL caching, sliding-window rate limiting, and persistent quota accounting, holding a 266-part board to roughly 281 calls against a 1,000-per-day cap.
- **FPGA Capture Validation Tool:** Self-initiated after scoping an engineer's manual waveform inspection as a data-comparison problem; reverse-engineered the vendor capture archive (ZIP, CSV, change-encoded VCD, JSON probe map), reconstructed the message stream from hardware handshake signals, and shipped offline Windows GUI and CLI executables reporting 13 named checks that locate corruption to the exact word and sample index.
- Investigated a check that contradicted known-good reference data rather than loosening it, identifying and documenting two previously undocumented hardware handshake behaviors; built an 11-scenario synthetic capture generator to prove the failure paths fire.

Mathematics Instructor & Tutor | Mr. J's Academy • Wyzant • Blue Train College Prep

Sept 2022 – May 2025 | Torrance & Goleta, CA

- Delivered 1,500+ hours of instruction from Algebra 1 through Calculus and SAT prep across 17+ recurring clients, holding a 4.8-star Wyzant rating and 100% retention; promoted to high school lead tutor with a 50% compensation increase. Diagnosed learning gaps from assessment data and built targeted problem sets, translating abstract quantitative material for non-technical audiences.

Lab Technician | Escape Communications

Summers 2017 – 2023, 2025 | Torrance, CA

- Performed QA testing and calibration on power supply units, recording multimeter readings into Excel trackers built to flag out-of-tolerance values automatically; tracked component inventory with low-count alerting and tested completed boards for compliance.

EDUCATION & CERTIFICATIONS

B.A. in Mathematics | University of California, Santa Barbara

Sept 2018 – March 2023

GPA 3.40 (Mathematics GPA 3.49) • 4x Dean's Honors • Minor: Secondary Mathematics & Science Education

Coursework: Probability & Statistics I and II, Linear Algebra, Differential Equations, Combinatorial Analysis, Methods of Analysis, Object Oriented Design and Data Structures (C++), Foundations of Computer Science, Microeconomics

Data Science Certificate | UCLA Extension

June 2026 – Dec 2026 (in progress)

Introduction to Data Science • Exploratory Data Analysis & Visualization • Big Data Management • Machine Learning Using Python

Google Data Analytics Professional Certificate | Coursera • 100% average across all nine courses

Oct 2025 – Jan 2026

Single Subject Mathematics Credential Coursework | CSU Long Beach • GPA 4.00

Aug 2024 – Oct 2025

SELECTED PORTFOLIO PROJECTS (15 projects at nadeaujonny.github.io)

- **Supply Chain Analytics** (Oracle SQL, Excel, Power BI): 180K orders; Power Query ETL, ABC/XYZ classification, ETS forecasting, MRP simulation, 6-page star-schema dashboard; diagnosed a structural 42.7% on-time delivery rate.
- **Skechers FP&A Model** (Excel): three-statement model from FY2020–FY2024 10-K filings; Bear/Base/Bull scenarios, sensitivity tables, KPI dashboard.
- **Customer Churn Prediction** (Python, scikit-learn): five classifiers on 7K records, SHAP-explained, deployed as a Streamlit scoring app at 0.83 AUC.